

Errata Sheet for:

Safety Equipment Reliability Handbook, Second Edition

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NOTE: The reliability data in the Safety Equipment Reliability Handbook is collected and published using best effort. The errata sheet will only address corrections to information that is identified as a misprint. It will not address revisions to data. All manufacturer and product specific data is provided for information only. The user should contact the manufacturer to check if any data updates are available.

Book corrections

ITEM	CORRECTION
1.5.41: Magnetrol Eclipse, 2nd generation	See detailed sheet on page 2.
1.6.14: MycroSENSOR 345 XTC	Equipment item should state "Prime Measurement 345 XTC". Manufacturer should state "PRIME Measurement Products, LLC".
2.41: Turck IM1-**Ex-R/T	Remarks should state: "1 or 2 inputs / 2 relay or transistor outputs".
2.42: Turck IM33-**Ex-Hi/24VDC	Remarks should state: "The two channels on the redundant boards shall not be used in the same safety function, e.g. to increase the hardware fault tolerance to achieve a higher SIL, as they contain common components".
4.1.7: ABB 800xA - Safety	See detailed sheet on page 3.
5.36: Turck IM35-**Ex-Hi/24VDC	Remarks should state: "The two channels on the redundant boards shall not be used in the same safety function, e.g. to increase the hardware fault tolerance to achieve a higher SIL, as they contain common components".

EQUIPMENT ITEM		Magnetrol Eclipse Enhanced Model 705			ITEM NO.		1.5.41	
GENERAL INFORMATION								
MANUFACTURER		Magnetrol International, Inc.						
MODEL		Eclipse Enhanced Model 705						
MEASUREMENT TYPE		Level - Guided Wave Radar						
ANALOG / DIGITAL		Analog			HARDWARE FAULT TOLERANCE		0	
ARCHITECTURE TYPE		B			CERTIFIED FOR USE UP TO SIL		N/A	
ASSESSMENT		FMEDA			BY	exida		
DATA SOURCE		FMEDA by exida						
REMARKS		None						
FAILURE RATE DATA			PER 10 ⁹ HOURS [FITs]					
			705-510X-XXX		705-51AX-XXX			
			OVERFILL / HIGH LEVEL	DRY-RUN / LOW LEVEL	OVERFILL / HIGH LEVEL	DRY-RUN / LOW LEVEL		
FAIL LOW			141	141	141	141		
FAIL HIGH			21	21	21	21		
FAIL DETECTED			405	405	488	488		
FAIL DANGEROUS DETECTED								
FAIL DANGEROUS UNDETECTED			183	183	106	106		
FAIL SAFE DETECTED								
FAIL SAFE UNDETECTED								
FAIL ANNUNCIATION DETECTED								
FAIL ANNUNCIATION UNDETECTED			38	38	31	31		
FAIL NO EFFECT			393	393	393	393		
SFF [%]			84.5	84.5	91.0	91.0		
APPLICATION EXAMPLE – OVERFILL, 705-510X-XXX								
PLC DETECTION BEHAVIOR		APPLICATION		λ^{SD}	λ^{SU}	λ^{DD}	λ^{DU}	SFF [%]
RANGE CHECKING	INPUT FILTERING	TRIP POINT	ALARM POINT					
ON	ON	HIGH	HIGH			567	183	84.5
ON	ON	HIGH	LOW			567	183	84.5
ON	OFF	HIGH	HIGH	426		141	183	84.5
ON	OFF	HIGH	LOW	21		546	183	84.5
OFF	OFF	HIGH	HIGH		426		324	72.6
OFF	OFF	HIGH	LOW		21		729	38.3

EQUIPMENT ITEM						ABB 800xA - Safety			ITEM NO.		4.1.7	
GENERAL INFORMATION												
MANUFACTURER		ABB Automation Technologies AB										
MODEL		800xA - Safety, Version 4.1										
LOGIC SOLVER TYPE		PLC				BETA FACTOR [%]				N/A		
CONFIGURATION		1001D				HARDWARE FAULT TOLERANCE				0		
ARCHITECTURE TYPE		B				CERTIFIED FOR USE UP TO SIL				2		
ASSESSMENT		IEC 61508 Certification				BY	TÜV Product Service GmbH					
DATA SOURCE		FMEDA by exida										
REMARKS		I/O Modules have internal hardware fault tolerance of 1 and are certified for use up to SIL3. Reliability data reflects 90% confidence and 20C.										
FAILURE RATE DATA				PER 10 ⁹ HOURS [FITs]								
	MODEL #	λ ^{SD}	λ ^{SU}	λ ^{DD}	λ ^{DU}	λ ^{AD}	λ ^{AU}	λ ^{NE}				
MAIN PROCESSOR	AC800M HI	0.00	5518	5234	0.84	286	773	7270				
POWER SUPPLY	SD82x, SS823	541	8	16	19	201	38	2660				
ANALOG IN MODULE	AI880 (8)	581	0.53	1534	0.64	511	4	1455				
ANALOG IN CHANNEL		5	0	131	0	12	0	104				
DIGITAL IN MODULE	DI880 (16)	294	16	726	0.64	749	47	1056				
DIGITAL IN CHANNEL		56	33	63	0	0.54	0	77				
ANALOG OUT MODULE												
ANALOG OUT CHANNEL												
DIGITAL OUT MODULE	DO880 (16)	1297	27	1090	0.75	824	45	1992				
DIGITAL OUT CHANNEL		5	0	0	0	201	42	128				
APPLICATION EXAMPLE												
The Safe Failure Fractions of the various configurations of this PLC support its use up to the SIL it has been certified for.												